

Exercise-induced oxidative stress: myths, realities and physiological relevance.

Although assays for the most popular markers of exercise-induced oxidative stress may experience methodological flaws, there is sufficient credible evidence to suggest that exercise is accompanied by an increased generation of free radicals, resulting in a measurable degree of oxidative modifications to various molecules. However, the mechanisms responsible are unclear. A common assumption that increased mitochondrial oxygen consumption leads per se to increased reactive oxygen species (ROS) production is not supported by in vitro and in vivo data. The specific contributions of other systems (xanthine oxidase, inflammation, haem protein auto-oxidation) are poorly characterised. It has been demonstrated that ROS have the capacity to contribute to the development of muscle fatigue in situ, but there is still a lack of convincing direct evidence that ROS impair exercise performance in vivo in humans. It remains unclear whether exercise-induced oxidative modifications have little significance, induce harmful oxidative damage, or are an integral part of redox regulation. It is clear that ROS play important roles in numerous physiological processes at rest; however, the detailed physiological functions of ROS in exercise remain to be elucidated.